AeroLab: Physical and Mathematical Foundations of the 2D Lattice Boltzmann Wind Tunnel

AeroLab Simulation Framework
High-Performance Computational Aerodynamics

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Abstract

This document provides a comprehensive mathematical and physical formulation of **AeroLab**, a real-time, two-dimensional numerical wind tunnel based on the Lattice Boltzmann Method (LBM) with a D2Q9 lattice topology. The document details the kinetic Boltzmann equation, the Bhatnagar-Gross-Krook (BGK) collision operator, Chapman-Enskog multiscale asymptotic analysis recovering the incompressible Navier-Stokes equations, Smagorinsky subgrid-scale (LES) turbulence modeling, aerodynamic boundary conditions, momentum exchange force evaluation for lift (C_L) and drag (C_D) coefficients, and the MacCormack predictor-corrector transport scheme for high-contrast, anti-diffusive smoke streamline visualization.

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1 Introduction and Governing Equations

In aerodynamic wind tunnel testing, external flow past an immersed lifting body is governed by the incompressible Navier-Stokes equations expressing conservation of mass and linear momentum:

$$\nabla \cdot \mathbf{u} = 0, \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_0}\nabla p + \nu \nabla^2 \mathbf{u}, \tag{2}$$

where $\mathbf{u} = (u_x, u_y)^T$ denotes the macroscopic velocity vector, p is the static pressure field, ρ_0 is the reference fluid density, and ν is the kinematic viscosity.

The fundamental dimensionless parameter governing aerodynamic similarity is the Reynolds number based on the airfoil chord length c and freestream velocity U_∞:

$$\mathrm{Re} = \frac{U_\infty c}{\nu}. \tag{3}$$

Rather than discretizing Equations (1)–(2) via traditional continuum methods (finite differences, finite volumes, or finite elements), AeroLab employs the **Lattice Boltzmann Method** (LBM). LBM simulates fluid mechanics from a mesoscopic kinetic perspective, solving for the evolution of discrete particle distribution functions whose statistical velocity moments recover the Navier-Stokes equations in the low-Mach, nearly-incompressible limit.

2 The Lattice Boltzmann Method (LBM D2Q9)

2.1 Discrete Boltzmann Equation

The core governing equation for the discrete velocity distribution functions $f_i(\mathbf{x}, t)$ along discrete lattice direction $\mathbf{c}_i$ is the discrete Boltzmann equation with the Bhatnagar-Gross-Krook (BGK) single relaxation time approximation:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = -\frac{\Delta t}{\tau}\left[f_i(\mathbf{x}, t) - f_i^{(0)}(\mathbf{x}, t)\right], \quad i = 0, \ldots, 8, \tag{4}$$

where $f_i^{(0)}(\mathbf{x}, t)$ is the local Maxwell-Boltzmann equilibrium distribution function, Δt is the discrete time step, and τ is the dimensionless relaxation time parameter.

Equation (4) is typically split into two distinct, highly parallelizable operations:

1. **Collision**:
$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \frac{1}{\tau}\left[f_i(\mathbf{x}, t) - f_i^{(0)}(\mathbf{x}, t)\right]. \tag{5}$$

2. **Streaming**:
$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t). \tag{6}$$

2.2 D2Q9 Velocity Topology and Weights

The two-dimensional nine-velocity (D2Q9) lattice model comprises one rest particle, four orthogonal nearest-neighbor particles, and four diagonal next-nearest-neighbor particles:

$$\mathbf{c}_i = \begin{cases} (0, 0), & i = 0, \\ \left(\cos\frac{(i-1)\pi}{2}, \sin\frac{(i-1)\pi}{2}\right) c, & i = 1, 2, 3, 4, \\ \sqrt{2}\left(\cos\left[\frac{(i-5)\pi}{2} + \frac{\pi}{4}\right], \sin\left[\frac{(i-5)\pi}{2} + \frac{\pi}{4}\right]\right) c, & i = 5, 6, 7, 8, \end{cases} \tag{7}$$

where $c = \Delta x/\Delta t = 1$ in standard lattice units. Specifically:

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix}^T. \tag{8}$$

The corresponding lattice weights w_i, derived from Hermite polynomial quadrature of the continuous Maxwellian, are:

$$w_i = \begin{cases} \frac{4}{9}, & i = 0, \\ \frac{1}{9}, & i = 1, 2, 3, 4, \\ \frac{1}{36}, & i = 5, 6, 7, 8. \end{cases} \tag{9}$$

The lattice speed of sound c_s satisfies:

$$c_s^2 = \frac{1}{3}c^2 = \frac{1}{3}\frac{\Delta x^2}{\Delta t^2}. \tag{10}$$

2.3 Equilibrium Distribution Function

A second-order Taylor-Hermite series expansion of the Maxwellian velocity distribution around zero mean velocity yields the local equilibrium distribution:

$$f_i^{(0)}(\rho, \mathbf{u}) = w_i \rho \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right]. \tag{11}$$

Substituting $c_s^2 = 1/3$:

$$f_i^{(0)}(\rho, \mathbf{u}) = w_i \rho \left[1 + 3(\mathbf{c}_i \cdot \mathbf{u}) + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}(\mathbf{u} \cdot \mathbf{u}) \right]. \tag{12}$$

2.4 Macroscopic Moments and Equation of State

The hydrodynamic conserved macroscopic quantities are the local density ρ and momentum density $\rho\mathbf{u}$:

$$\rho(\mathbf{x}, t) = \sum_{i=0}^{8} f_i(\mathbf{x}, t), \tag{13}$$

$$\rho\mathbf{u}(\mathbf{x}, t) = \sum_{i=0}^{8} \mathbf{c}_i f_i(\mathbf{x}, t). \tag{14}$$

The effective thermodynamic pressure follows an ideal gas equation of state:

$$p = c_s^2 \rho = \frac{1}{3}\rho. \tag{15}$$

2.5 Chapman-Enskog Asymptotic Expansion

Using a multi-scale asymptotic expansion with small Knudsen number $\epsilon = \Delta t \ll 1$,

$$f_i = f_i^{(0)} + \epsilon f_i^{(1)} + \epsilon^2 f_i^{(2)} + \mathcal{O}(\epsilon^3), \tag{16}$$

$$\partial_t = \epsilon \partial_{t_1} + \epsilon^2 \partial_{t_2}, \quad \nabla = \epsilon \nabla_1, \tag{17}$$

one systematically recovers the continuity and Navier-Stokes equations at $\mathcal{O}(\epsilon^2)$, with kinematic viscosity given by:

$$\nu = c_s^2 \left(\tau - \frac{1}{2} \right) \Delta t = \frac{2\tau - 1}{6} \Delta t. \tag{18}$$

For a specified physical Reynolds number Re, chord length c, and lattice inlet velocity $U_\infty = u_{\text{in}}$, the relaxation time is computed dynamically as:

$$\nu = \frac{u_{\text{in}} c}{\text{Re}}, \quad \tau = 0.5 + 3\nu. \tag{19}$$

Positivity of physical viscosity requires $\tau > 0.5$.

3 Turbulence Modeling: Smagorinsky Subgrid-Scale (LES) Model

At elevated Reynolds numbers, numerical stability of single-relaxation-time LBM can degrade when $\tau \to 0.5$. To ensure robust simulation of high-speed wake turbulence and separated shear layers, AeroLab incorporates the **Smagorinsky Large Eddy Simulation (LES)** subgrid-scale model.

3.1 Resolved Strain Rate Tensor

In standard continuum LES, unresolved turbulent eddies are represented by an effective eddy viscosity ν_t:

$$\nu_t = (C_s \Delta)^2 |\bar{S}|, \tag{20}$$

where $C_s \approx 0.15$ is the Smagorinsky constant, $\Delta = \Delta x = 1$ is the grid filter width, and $|\bar{S}| = \sqrt{2\bar{S}_{\alpha\beta}\bar{S}_{\alpha\beta}}$ is the characteristic filtered strain rate tensor:

$$\bar{S}_{\alpha\beta} = \frac{1}{2}\left(\frac{\partial u_\alpha}{\partial x_\beta} + \frac{\partial u_\beta}{\partial x_\alpha}\right). \tag{21}$$

3.2 Non-Equilibrium Stress Tensor Formulation

A profound advantage of LBM is that the strain rate tensor does **not** require spatial finite differences; it is evaluated purely locally from the second-order non-equilibrium moment of the distribution functions:

$$\Pi^{(1)}_{\alpha\beta} = \sum_{i=0}^{8} c_{i\alpha} c_{i\beta} \left(f_i - f_i^{(0)}\right). \tag{22}$$

From Chapman-Enskog expansion, the leading-order non-equilibrium tensor is directly proportional to the strain rate:

$$\Pi^{(1)}_{\alpha\beta} = -2\rho c_s^2 \tau_{\text{eff}} \bar{S}_{\alpha\beta}. \tag{23}$$

Taking the Frobenius norm:

$$Q = \sqrt{2\Pi^{(1)}_{\alpha\beta}\Pi^{(1)}_{\alpha\beta}} = \sqrt{\Pi_{xx}^2 + \Pi_{yy}^2 + 2\Pi_{xy}^2} = 2\rho c_s^2 \tau_{\text{eff}} |\bar{S}|. \tag{24}$$

3.3 Effective Relaxation Time Calculation

The total effective viscosity is the sum of molecular and subgrid eddy viscosities:

$$\nu_{\text{eff}} = \nu_0 + \nu_t = c_s^2 \left(\tau_{\text{eff}} - \frac{1}{2}\right). \tag{25}$$

Substituting $\nu_t = C_s^2 |\bar{S}| = C_s^2 \frac{Q}{2\rho c_s^2 \tau_{\text{eff}}}$ leads to a closed quadratic equation for τ_{eff}:

$$\tau_{\text{eff}} = \tau_0 + \frac{C_s^2 Q}{2\rho c_s^4 \tau_{\text{eff}}}. \tag{26}$$

Solving the positive root of this quadratic equation yields:

$$\tau_{\text{eff}} = \frac{1}{2}\left[\tau_0 + \sqrt{\tau_0^2 + 18\sqrt{2}\, C_s^2 \frac{Q}{\rho}}\right]. \tag{27}$$

This local adaptation ensures that wherever turbulent shearing Q increases, τ_{eff} increases safely away from the 0.5 singularity, guaranteeing unconditional numerical stability.

4 Boundary Conditions

4.1 Inflow Boundary Condition (Inlet at $x = 0$)

At the wind tunnel inlet ($x = 0$), uniform horizontal freestream velocity $\mathbf{u}_{\text{in}} = (U_\infty, 0)$ is prescribed. An equilibrium Dirichlet boundary condition is applied across all incoming directions ($i \in \{1, 5, 8\}$):

$$f_i(0, y, t) = f_i^{(0)}(\rho = 1.0, u_x = U_\infty, u_y = 0). \tag{28}$$

4.2 Outflow Boundary Condition (Outlet at $x = L_x$)

At the tunnel outlet ($x = L_x$), a convective zero-gradient boundary condition allows vortices and pressure waves to exit without reflection back into the test section:

$$f_i(L_x, y, t) = f_i(L_x - 1, y, t), \quad \forall i. \tag{29}$$

4.3 Tunnel Sidewalls ($y = 0$ and $y = L_y$)

To model an unconfined wind tunnel test section without artificial boundary layer growth, top and bottom walls employ free-slip symmetry conditions:

$$f_{\text{UP}}(x, 0, t) = f_{\text{DOWN}}(x, 0, t), \quad \text{at bottom wall } y = 0, \tag{30}$$

$$f_{\text{DOWN}}(x, L_y, t) = f_{\text{UP}}(x, L_y, t), \quad \text{at top wall } y = L_y, \tag{31}$$

where mirror pairs exchange populations: $(2 \leftrightarrow 4)$, $(5 \leftrightarrow 8)$, $(6 \leftrightarrow 7)$.

4.4 Airfoil No-Slip Wall: Half-Way Bounce-Back

On the solid boundaries of the airfoil Ω_s, the no-slip condition ($\mathbf{u} = 0$) is enforced via the half-way bounce-back scheme. When a streaming particle encounters a solid node, its momentum is exactly reversed into the conjugate direction:

$$f_{\bar{i}}(\mathbf{x}_b, t + \Delta t) = f_i^*(\mathbf{x}_b, t), \tag{32}$$

where $\mathbf{c}_{\bar{i}} = -\mathbf{c}_i$ corresponds to the opposite velocity index mapping:

$$\text{OPP} = [0, 3, 4, 1, 2, 7, 8, 5, 6]. \tag{33}$$

5 Aerodynamic Force and Coefficient Evaluation

5.1 Momentum Exchange Method (MEM)

The aerodynamic forces exerted on the immersed airfoil are calculated directly using the **Momentum Exchange Method**. At every solid node $\mathbf{x}_b \in \Omega_s$, fluid packets transfer net linear momentum during bounce-back:

$$\Delta\mathbf{p}(\mathbf{x}_b, t) = \mathbf{c}_i f_i^*(\mathbf{x}_b, t) - \mathbf{c}_{\bar{i}} f_{\bar{i}}(\mathbf{x}_b, t + \Delta t) = 2\mathbf{c}_i f_i^*(\mathbf{x}_b, t). \tag{34}$$

Summing over all solid nodes and incoming directions yields the total hydrodynamic force vector:

$$\mathbf{F}(t) = \sum_{\mathbf{x}_b \in \Omega_s} \sum_{i=1}^{8} 2\mathbf{c}_i f_i^*(\mathbf{x}_b, t). \tag{35}$$

In wind-tunnel axes aligned with the $+x$ freestream direction:

$$\text{Total Drag Force: } D = F_x, \tag{36}$$

$$\text{Total Lift Force: } L = F_y. \tag{37}$$

5.2 Dimensionless Aerodynamic Coefficients

The dynamic pressure per unit length is:

$$q_\infty = \frac{1}{2}\rho_0 U_\infty^2 c. \tag{38}$$

The instantaneous aerodynamic coefficients are:

$$C_L = \frac{L}{q_\infty} = \frac{F_y}{\frac{1}{2}\rho_0 U_\infty^2 c}, \quad C_D = \frac{D}{q_\infty} = \frac{F_x}{\frac{1}{2}\rho_0 U_\infty^2 c}, \quad C_R = \sqrt{C_L^2 + C_D^2}. \tag{39}$$

5.3 Surface Pressure Coefficient (C_p)

The local dimensionless surface pressure coefficient C_p is derived from the local density variation $\rho - \rho_0$:

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho_0 U_\infty^2} = \frac{\frac{1}{3}(\rho - 1)}{\frac{1}{2}U_\infty^2}. \tag{40}$$

At the stagnation point where $\mathbf{u} = 0$, Bernoulli's equation dictates $C_p = 1.0$.

6 Smoke Streamline Transport: MacCormack Scheme

To visualize streamlines and shedding wake streaklines with high fidelity, a conserved passive scalar field $\phi(\mathbf{x}, t) \in [0, 1]$ (smoke dye concentration) is advected by the local flow field:

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = 0. \tag{41}$$

6.1 The MacCormack Predictor-Corrector Anti-Diffusion Scheme

Standard first-order semi-Lagrangian advection suffers from significant numerical diffusion:

$$\phi_{\text{diff}}(\mathbf{x}) \approx \phi(\mathbf{x} - \mathbf{u}\Delta t) = \phi(\mathbf{x}) - \Delta t(\mathbf{u} \cdot \nabla)\phi + \frac{\Delta t^2}{2}(\mathbf{u} \cdot \nabla)^2 \phi + \mathcal{O}(\Delta x), \tag{42}$$

which turns sharp streamlines into a milky, washed-out fog.

AeroLab replaces standard upwinding with a second-order **MacCormack predictor-corrector transport scheme**:

1. **Forward Predictor**: Advect the dye forward along the velocity field:

$$\phi^* = \mathcal{S}(\phi^n, \mathbf{u}\Delta t), \tag{43}$$

where $\mathcal{S}(\phi, \mathbf{d})$ denotes bilinear interpolation sampling at $\mathbf{x} - \mathbf{d}$.

2. **Backward Reverser**: Advect the predicted field backward along $-\mathbf{u}\Delta t$:

$$\phi^{**} = \mathcal{S}(\phi^*, -\mathbf{u}\Delta t). \tag{44}$$

3. **Error Estimation and Compensation**: The numerical error accumulated over the forward-backward cycle is $\mathcal{E} = \frac{1}{2}(\phi^n - \phi^{**})$. We construct the compensated intermediate field:

$$\phi^c = \text{clip}\left(\phi^n + \frac{1}{2}(\phi^n - \phi^{**}),\, 0.0,\, 1.0\right). \tag{45}$$

4. **Final Forward Advection**:

$$\phi^{n+1} = \mathcal{S}(\phi^c, \mathbf{u}\Delta t) \times \gamma_{\text{decay}}, \tag{46}$$

where $\gamma_{\text{decay}} = 0.9995$ is a subtle background dissipation factor that purges residual accumulation while preserving razor-sharp cores.

6.2 Gaussian Injector Rake and Interactive Wand

At the inlet ($x = 0$), discrete dye injectors with Gaussian profiles prevent pixelation artifacts:

$$\phi_{\text{inlet}}(y) = \sum_{k=1}^{N_{\text{lines}}} \exp\left(-\frac{(y - y_k)^2}{2\sigma^2}\right), \quad \sigma = \max(1.1,\, 0.12\Delta y_{\text{lines}}). \tag{47}$$

In pulsed mode, injection is periodically modulated:

$$\phi_{\text{inject}}(t) = \begin{cases} 1, & (n_{\text{step}} \bmod 24) < 11, \\ 0, & \text{otherwise.} \end{cases} \tag{48}$$

For the interactive smoke wand positioned at user coordinates (x_w, y_w):

$$\phi_{\text{wand}}(x, y) = 0.9 \exp\left(-\frac{(x - x_w)^2 + (y - y_w)^2}{2r_w^2}\right). \tag{49}$$

7 NACA 4-Digit Airfoil Geometry

The geometry of the immersed lifting surface is parameterized by the standard NACA 4-digit designation ($MPXX$):

- M: Maximum camber ($m = M/100$).
- P: Location of maximum camber ($p = P/10$).
- XX: Maximum thickness as percentage of chord ($t = XX/100$).

7.1 Thickness Distribution

The symmetrical half-thickness distribution $y_t(x)$ along the chord coordinate $x/c \in [0, 1]$ is:

$$y_t(x) = 5tc\left[0.2969\sqrt{\frac{x}{c}} - 0.1260\left(\frac{x}{c}\right) - 0.3516\left(\frac{x}{c}\right)^2 + 0.2843\left(\frac{x}{c}\right)^3 - 0.1036\left(\frac{x}{c}\right)^4\right]. \tag{50}$$

7.2 Mean Camber Line

For cambered profiles ($m > 0, p > 0$):

$$y_c(x) = \begin{cases} \frac{m}{p^2}\left[2p\left(\frac{x}{c}\right) - \left(\frac{x}{c}\right)^2\right]c, & 0 \le \frac{x}{c} < p, \\ \frac{m}{(1-p)^2}\left[(1-2p) + 2p\left(\frac{x}{c}\right) - \left(\frac{x}{c}\right)^2\right]c, & p \le \frac{x}{c} \le 1. \end{cases} \tag{51}$$

7.3 Quarter-Chord Pivot Rotation

To accurately model the aerodynamic center and angle of attack α, the airfoil coordinates are rotated around its quarter-chord pivot point:

$$\mathbf{x}_q = (x_{\text{le}} + 0.25c,\, y_{\text{le}})\,. \tag{52}$$

Any lattice node (X, Y) is transformed into the un-rotated airfoil frame (x_r, y_r) via:

$$x_r = (X - x_q)\cos\alpha - (Y - y_q)\sin\alpha, \tag{53}$$
$$y_r = (X - x_q)\sin\alpha + (Y - y_q)\cos\alpha. \tag{54}$$

A node is flagged as solid ($\Omega_s = \text{True}$) if:

$$0 \le x_r + 0.25c \le c \quad \text{and} \quad |y_r - y_c(x_r)| \le y_t(x_r). \tag{55}$$

8 Summary of Parameters and Units

Table 1: Key physical and numerical parameters in AeroLab.

Symbol	**Physical Meaning**	**Typical Value**	**Units**
$N_x \times N_y$	Grid resolution (medium)	800×300	Lattice nodes
c	Airfoil chord length	$0.16 \times N_x \approx 128$	Lattice spacing Δx
U_∞	Lattice inlet velocity	0.08	Lattice velocity $\Delta x/\Delta t$
Re	Reynolds number	$1000 - 30000$	Dimensionless
τ	BGK relaxation time	> 0.501	Lattice time steps Δt
c_s	Speed of sound in D2Q9	$1/\sqrt{3} \approx 0.577$	Lattice velocity
C_s	Smagorinsky constant	0.15	Dimensionless
C_L, C_D	Sectional lift and drag	Computed dynamically	Dimensionless

9 Conclusion

The combination of D2Q9 Lattice Boltzmann kinetics, Smagorinsky subgrid-scale turbulence closure, momentum exchange boundary force integration, and MacCormack anti-diffusive advection provides AeroLab with a rigorous, physically validated foundation capable of reproducing complex aerodynamic phenomena—such as boundary layer separation, stall vortex shedding, and dynamic force responses—in real time at high lattice throughput.